Playbook: Graph Construction, Review, and Causal Attribution

A Decision Guide for the Causal Modeling Pipeline

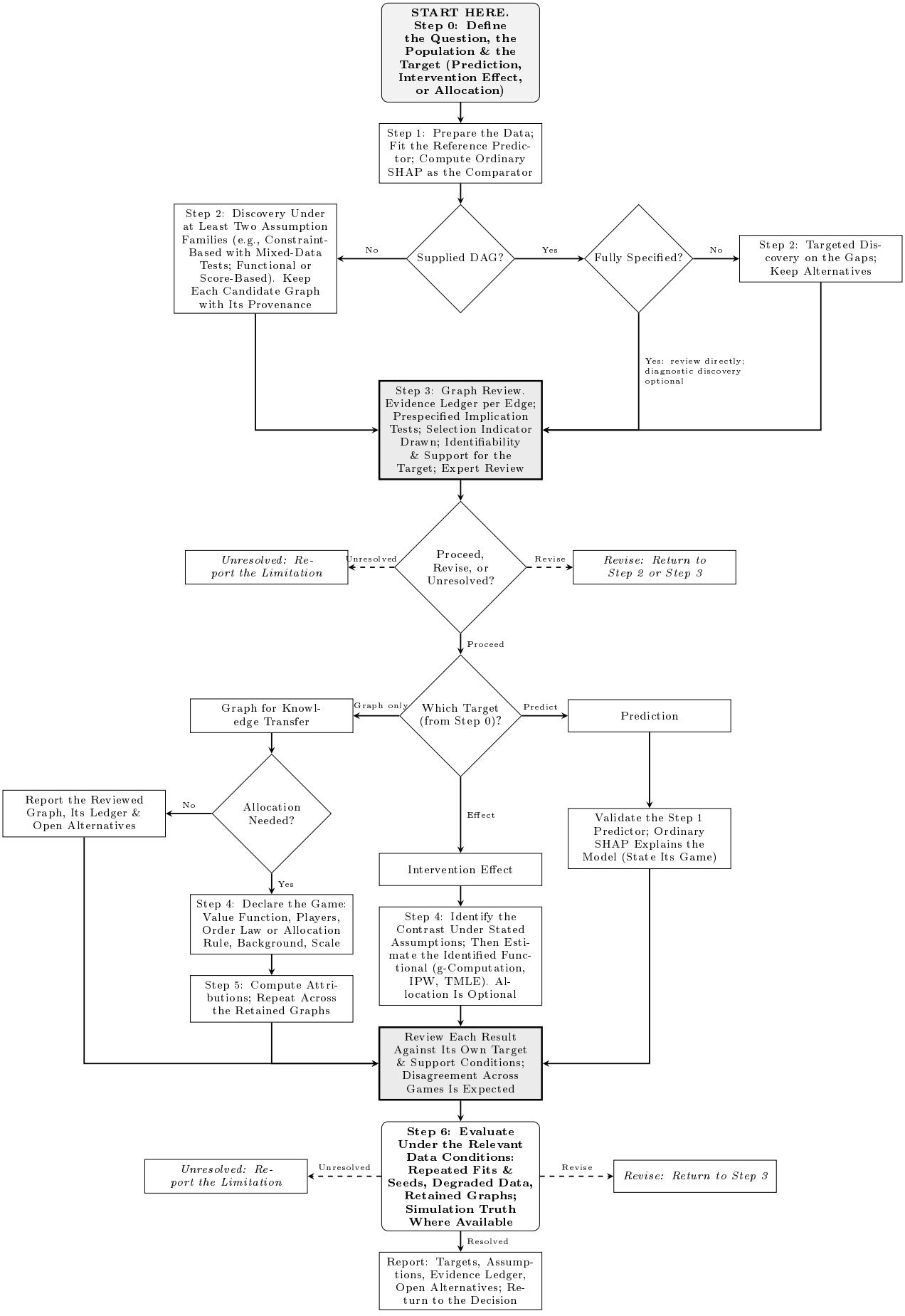
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Scope & Prerequisites

Scope: Assumes a dataset in which the variables of interest are measured and in which selection into the dataset can be represented. Dataset identification and raw ETL are out of scope. The steps are the manuscript's Steps 0–6.
Core principle: No single analysis establishes a DAG or an attribution. Confidence in a graph comes from its testable implications and from evidence about its edges gathered outside the dataset. It also depends on how much the target changes across the alternatives still open. Agreement between algorithms run on one dataset can be limited robustness evidence when their assumptions differ; it is not independent validation of the graph. Every branch ends in expert review.
Status of these recommendations: They rest on reasoning and the cited literature. The accompanying simulation study tested none of them except where a record is named in the text.

Key Execution Rules & Safeguards

1. **Triangulate; Do Not Duplicate:** Pair a method with one whose main sources of bias differ, or with evidence from outside the dataset (experiments, timing, mechanism) [1, 2]. Agreement between algorithms run on the same data cannot detect shared measurement, selection or omitted-variable error, and benchmark success can come from a shared artifact such as variance ordering [3]. Where aggregation is wanted, use a defined ensemble or validation procedure rather than a vote [4, 5].
2. **Report Structural Uncertainty; Do Not Merge Graphs by Hand:** Keep each candidate graph or equivalence class with its provenance. A set union of outputs need not be acyclic or orientation-consistent, and an intersection cannot raise recall; it only removes edges. Markov-and-faithfulness methods such as PC and GES return an equivalence class, so report the set of effects compatible with it [6, 7]; functional methods such as LiNGAM can identify a DAG under stronger assumptions [8]. Edge frequencies from resampling measure the stability of the fitted procedure, not causal confidence [9, 10]. Match tests to the variable types [11, 12]; the renal Gaussian-PC run that found no outcome adjacency is a mixed-data diagnostic awaiting follow-up.
3. **Test Implications; Rejection Is the Informative Outcome:** The conditional independencies a graph implies are its testable content [13, 14]. Prespecify them and test on held-out or external data where possible. Rejection challenges the graph or the test's assumptions; non-rejection is not confirmation. Report multiplicity, the test's assumptions and power against relevant departures: no conditional-independence test is uniformly valid without assumptions [15], near-faithfulness violations and finite power make weak dependencies hard to detect [16, 17], and tests run on the data that selected the graph need post-selection handling [18].
4. **Represent Selection Explicitly:** Add a selection indicator with arrows from its causes and represent the analyzed cohort as conditioned on that indicator. Some implications of a correct DAG then fail for that reason alone [19].
5. **Positivity Belongs to the Target, Not to Node Depth:** Positivity is defined for a specified intervention, population and required support [20, 21]. Where support is absent the contrast is not identified; where data are sparse the estimate is imprecise; predictive SHAP stays computable for a rare feature but is unstable. Report support diagnostics for each method and estimand.
6. **Declare the Game Before Reading the Number:** An attribution answers the question its value function asks: conditional, model-interventional, outcome-based do-Shapley, or a fitted-model interventional analogue, each with its players, order law or allocation rule, background and scale [22–26]. In this study's matched comparison, ordering alone produced no detected improvement. The oracle-mechanism prototype changed both the value function and the order law; its larger agreement is provisional and does not isolate which change caused it.
7. **Causal Inputs Carry Their Errors Forward:** Each family consumes different causal information: an ordering for asymmetric Shapley values [25], edge structure for Shapley Flow [26], identified interventional distributions or fitted conditionals for causal Shapley and do-Shapley [23, 24], and supplied mechanisms for the propagation prototype. Errors in those inputs enter the attributions, so report results across the retained graphs. No claim about small samples is made until the degradation experiment runs.
8. **Expert Review Closes Every Branch:** This is the paper's proposed governance rule. The domain team reviews structure and results together, keeps a ledger of what changed and why, and returns unresolved disagreement to Step 3. NASA's diagramming practice supplies the configuration-management precedent [27] and its evidence levels grade the edges [28, 29]. Where a simulation truth exists, Step 6 evaluates performance against it; success against a simulation oracle does not validate the graph for astronauts.



Shaded boxes with heavy borders are review steps. Step numbers are the manuscript's Steps 0–6. The prediction route does not require Steps 4–5; an effect-only analysis can omit Step 5.

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